

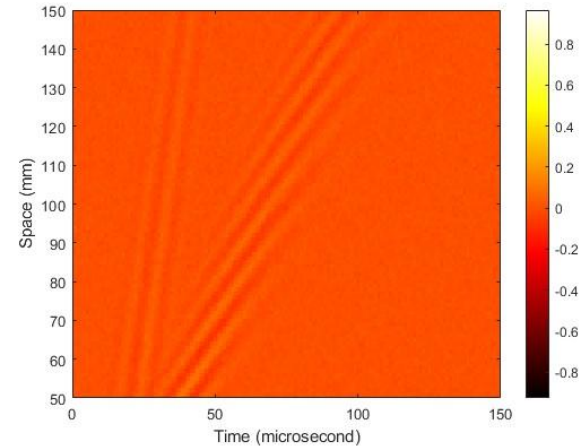
Machine Learning for Material Characterization Using Wave Propagation Data

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Problem

Material changes are difficult to identify early

- ❑ Changes or damage can alter how waves travel through a material.
- ❑ These changes may not be visible through visual inspection.
- ❑ Manual interpretation of wave data can be difficult and time-consuming.
- ❑ A simple, reliable method is needed to identify material changes from wave measurements.

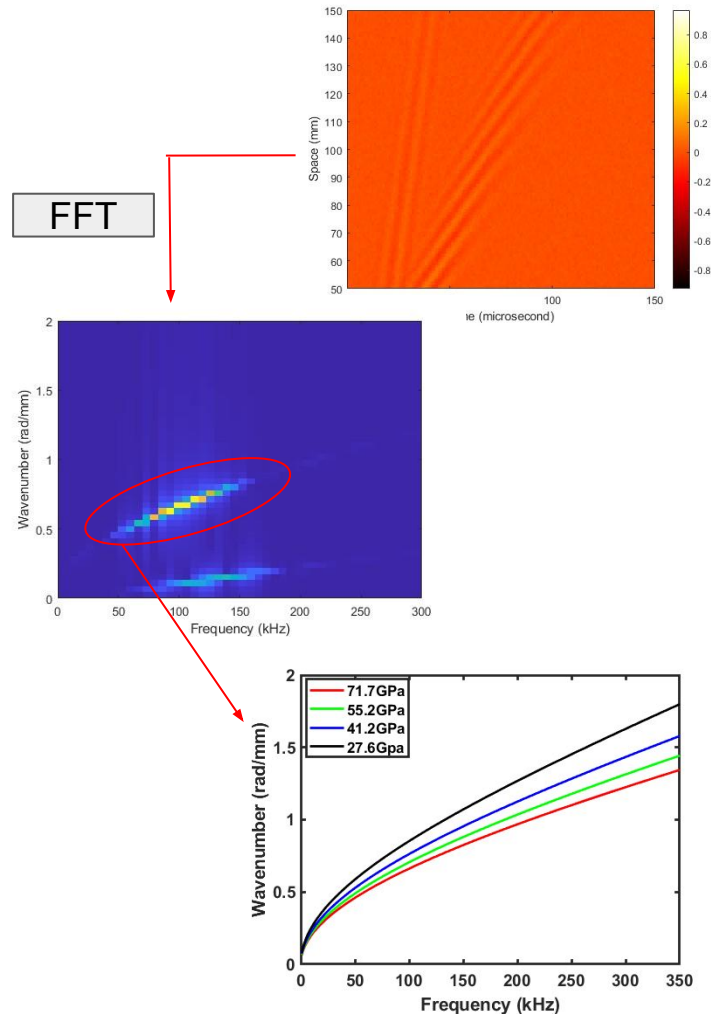


Proposed solution

Use machine learning to analyze wave patterns

- ❑ Generate 1D wave-propagation data using an analytical model.
- ❑ Represent the data in:
 - ❑ time–space form, showing wave motion over time;
 - ❑ frequency–wavenumber form, showing changes in wave patterns.
- ❑ Train a machine-learning model to recognize shifts in the wave pattern.
- ❑ Use the detected shift to classify or estimate changes in material properties.

FFT



Evaluation Plan

Test whether ML can identify material changes

- ❑ Create simulated data for multiple material-property conditions.
- ❑ Split the data into training and test sets.
- ❑ Compare predicted material conditions with known analytical-model inputs.
- ❑ Evaluate performance using classification accuracy or prediction error.
- ❑ Assess whether time–space data improves detection compared with frequency–wavenumber data.

Thank You

Questions?